Field observation — concrete structure near the 1948 node

Ladera Ranch vicinity, Orange County, California · recorded 2026-08-07 · LHDRS field record

THIS IS NOT AN IDENTIFICATION.

The structure below has not been measured to a standard, has not been sampled, and has not been examined by anyone qualified to identify it. Dimensions are as reported by the observer and are approximate. This document exists to record what was seen, set out what would confirm or exclude a cattle-dipping vat, and be handed to someone able to decide.

SAFETY — READ BEFORE RETURNING TO THE SITE

If this *is* a dipping vat, the sediment inside it is the single most concentrated place arsenic would be found anywhere in this landscape — this project's own modelling puts that compartment in the thousands of mg/kg against a California background of 1–11 mg/kg. **Do not dig in it. Do not handle or remove the sediment. Do not let children or animals into it.** Photograph and measure from outside. Arsenic does not degrade; a century has not reduced it.

The structure and its surroundings, as found

Photographs exactly as taken — no rotation, cropping or enhancement. Archived with SHA-256 checksums in evidence/lhdrs/field_observations/.

1 — Axis view. The concrete channel runs away into brush; pipe frame at the near end.

2 — End view. Clean 90° concrete corner, raised rim on both sides, pipe railing descending INTO the structure. The strongest visual argument for a discrete vessel.

3 — Weathered timber post with WIRE still attached, standing near the structure; a second short stake beside it. Utility pole in the background.

4 — Fallen and leaning pipe rails in the brush beside the structure — consistent with chute or pen railing.

What was observed

Attribute	As reported	Confidence
Material	Concrete	observer, direct
Length	REVISED: approximately 12 ft (earlier estimate of 15–20 ft superseded on second look)	estimated, not measured
Width	approximately 3–4 ft	estimated, not measured
Depth	NOT MEASURED — obscured by infill	unknown
Form	Discrete linear channel with ends; not a continuous ditch run	observer, direct
Rim & corner	Raised concrete rim on both long sides; clean 90-degree corner at the near end	photo 2
Railing	Rusted pipe frame at the near end, one rail descending INTO the structure; further fallen/leaning pipe rails in the brush beside it	photos 1, 2, 4
Timber post	Weathered squared timber post standing nearby with WIRE still attached; a second short stake beside it	photo 3
Wire	Strand wire at the post, consistent with pen/fence line	photo 3

Context	Utility pole visible in background of photo 3 (locational aid); dry mustard/brush cover throughout	photos
Setting	Heavy dry brush, drainage corridor, near the 1948 mapped structure	observer, direct
Coordinates	NOT RECORDED — no GPS in the image metadata	missing
Image	640 x 480 px, no EXIF GPS or timestamp retained	low resolution

The federal specification for a concrete dipping vat

USDA Bureau of Animal Industry Circulars 183 (1911) and 207 (1912) give construction specifications for **concrete** cattle-dipping vats. Circular 174 (1911) states cement is *preferable* to lumber because it "has not the disadvantage of leaking, which is common in wooden vats." **Concrete construction is therefore consistent with, and in fact the recommended form of, a period dipping vat.**

Dimension	Circular 183 / 207 concrete vat	Structure as reported
Length	26 ft at top, tapering to 12 ft at bottom	~12 ft (revised) — matches the FLOOR length
Width	3.0 ft at top, tapering to 1.5 ft at bottom	~3–4 ft
Depth	6.5 ft	not measured
Capacity	1,470 gal at 5 ft 3 in fill	unknown
Chute	30 in wide, 20 ft long, leading in	not observed
Drip pen	~36 in wide, 20–40 ft, sloped back toward vat	not observed

Why it could be a vat — the case for

- **Concrete is the specified material.** Not merely compatible — Circular 174 recommends it over timber specifically because timber vats leaked.
- **Revised length matches the spec floor exactly.** The documented vat is 12 ft at the floor widening to 26 ft at the rim. The observer's revised estimate of ~12 ft equals the floor length — consistent with the visible concrete being the base course of a full-size vat, with the flaring upper walls buried, broken, or (in a hybrid build) never concrete at all.
- **Reported width brackets the spec width.** 3–4 ft observed; the spec runs 1.5 ft at the floor to 3.0 ft at the rim, so the reading sits at/above the rim width — widths were estimates.
- **It is a discrete structure with ends.** A concrete-lined drainage channel runs continuously across a slope; it does not begin and end in 15–20 ft. A vat does.
- **A metal frame is present.** Vats were fitted with rails, splash boards on hinged posts, and chute fencing. A rusted pipe frame at one end is consistent with that, though not diagnostic.
- **The setting fits the siting logic.** Vats were placed near water and stock-handling ground. The structure sits in a drainage corridor near the only mapped 1948 structure in the study area.

Why it might not be — the case against

Stated at equal length, because a document that only argues one way is worth nothing to the agency or specialist who has to act on it.

- **Depth is unmeasured, and depth is the whole question.** A dipping vat is **6.5 ft deep** with steep sides — cattle swim through it. If this structure proves to be 1–2 ft deep it is a trough, a flume or a lined ditch, and nothing else about it matters.
- **Concrete-lined drainage is extremely common here.** Brow ditches, V-ditches and terrace drains with pipe railings are standard on California graded slopes and in flood-control corridors. Many are 3–4 ft wide.
- **No taper has been confirmed.** A vat narrows markedly from rim to floor. A ditch has parallel walls. This has not been checked.
- **Neither end feature has been observed.** A vat is asymmetric — a steep entry slide at roughly 25° at one end, a cleated exit ramp at the other. Their absence would be strong evidence against.
- **No drip pen or chute has been seen.** These are large, concrete, and adjacent. Their absence is not conclusive — they may be buried or demolished — but their presence would be near decisive.
- **The systematic aerial search found no vat in this area.** 37,228 cells across the 1929, 1938 and 1947 frames returned nothing vat-like inside the study frame. That is weak evidence of absence, but it is on the record and must be weighed.
- **Age is unestablished.** Nothing observed dates the concrete. It could postdate the dipping era by decades.

Why a lone concrete channel is what a whole station leaves behind

A dipping vat was never a standalone object. Circular 183 specifies a complete **station**: receiving and retaining pens, a chute, the vat itself, an exit incline, a dripping pen, and a barrel sunk in the ground to catch and recycle run-off. Cattle moved through it in one direction.

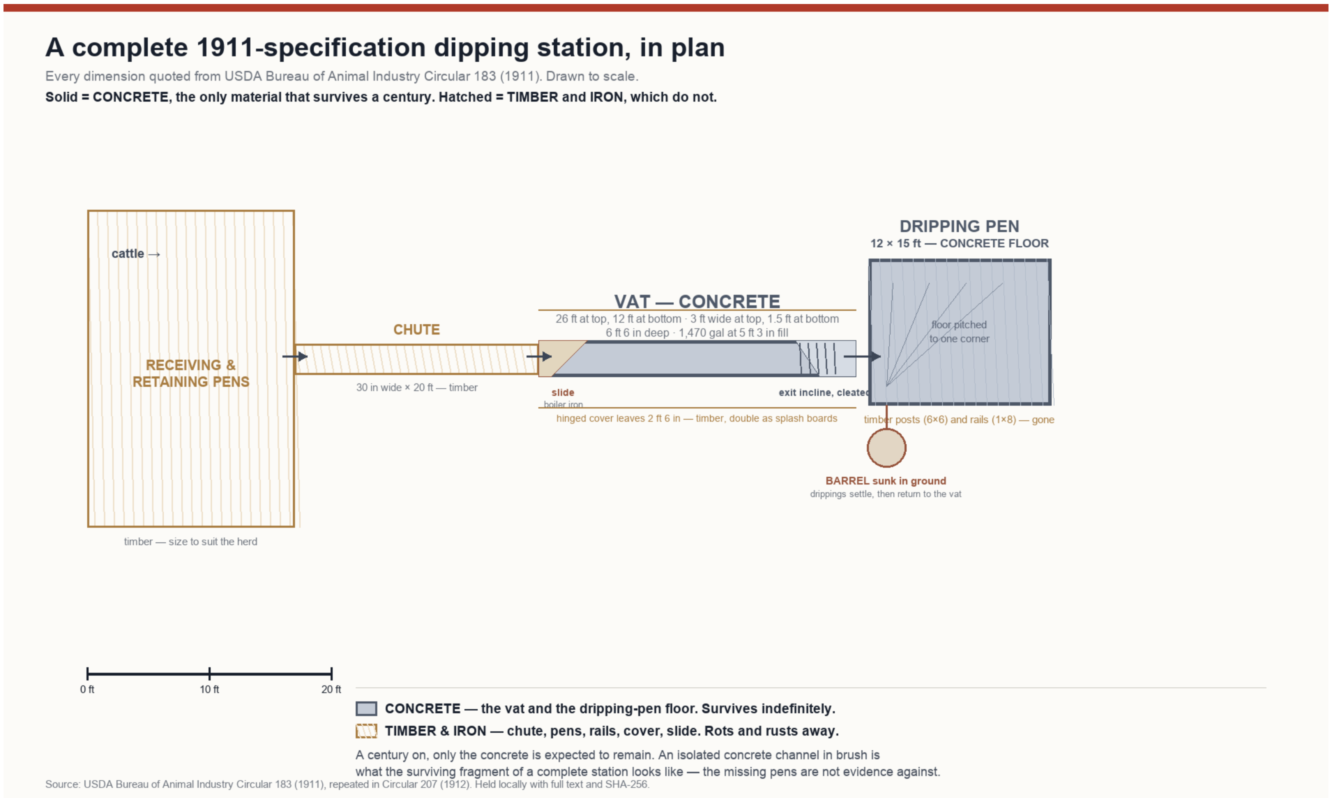
The decisive detail is in the bill of materials. **The station splits cleanly by material, and only two components were concrete.**

Component	Specified material	Expected after a century
Vat	CONCRETE (or timber; concrete preferred)	SURVIVES
Dripping-pen floor	CONCRETE, 12 × 15 ft, pitched to a corner	SURVIVES
Chute, 30 in × 20 ft	timber	gone
Receiving / retaining pens	timber	gone
Dripping-pen posts and rails	timber, 6×6 posts, 1×8 rails	gone
Cover leaves / splash boards	timber on posts set 3 ft deep	gone
Entry slide facing	sheet boiler iron on the cement	rusted away
Drainage pipe and barrel	iron pipe, barrel sunk in ground	rusted / buried

What follows from that

A century on, the timber has rotted and the ironwork has largely gone. **The concrete is the only part expected to remain by default.** So an isolated concrete channel sitting in brush, with no pens, no chute and no fencing around it, is *precisely* what the surviving fragment of a complete dipping station would look like. The absence of the other components is the predicted condition — not evidence against.

This cuts the other way too, and should be said: the same argument means a surviving concrete fragment carries **less** corroborating context than an intact site would. There is no chute or pen left to confirm the interpretation. That is why the measurements in the next section carry so much weight — the structure has to identify itself.



Scale plan of a complete 1911-specification station, drawn from the federal instructions. Every dimension is quoted from USDA Bureau of Animal Industry Circular 183 (1911), repeated in Circular 207 (1912). Solid = concrete; hatched = timber and iron.

The revised measurement, and what sat on top

On a second look the observer revised the length estimate from 15–20 ft down to **approximately 12 ft**. Against the federal specification this is the most significant measurement yet, because **12 ft is exactly the specified FLOOR length of the vat** (26 ft at the rim tapering to 12 ft at the floor).

Observed length	Height above vat floor (from the taper)
~12 ft	0 ft — THE FLOOR ITSELF
13 ft	0.5 ft
14 ft	0.9 ft
20 ft	3.7 ft
26 ft	6.5 ft — the original rim

An observed ~12-ft concrete rectangle is therefore consistent with the visible concrete being the **floor pan and lowest wall course** of a full-size vat — the base — with whatever stood above it now gone. Two documented construction variants explain the missing top, and they answer the question directly:

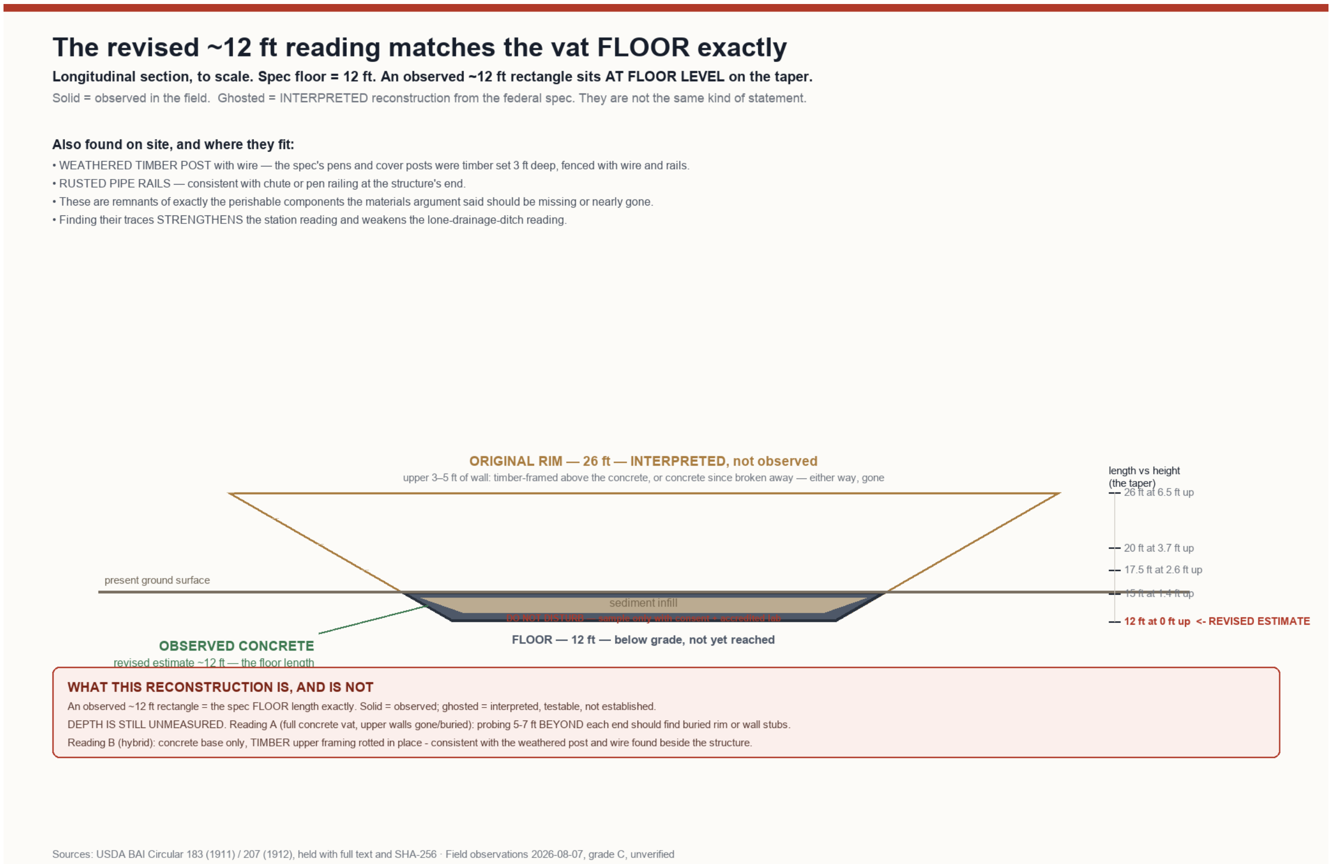
READING A — the top was CONCRETE, and it is either buried or broken

Circulars 183/207 describe a full-depth concrete vat cast in the excavation, walls flaring from 12 ft at the floor to 26 ft at the rim. If this was that type, the upper walls did not vanish — they are either (a) **BURIED**: a century of slopewash and sediment can bury the flaring upper walls outward and below present grade, leaving only the innermost course exposed; or (b) **BROKEN**: unreinforced upper courses crack from roots, frost and stock traffic and are pushed over — in which case rubble fragments should lie in the brush within a few yards. **Testable without digging: probe the soil 5–7 ft BEYOND each visible end. A buried rim or wall stub found there confirms Reading A.**

READING B — the top was TIMBER, above a concrete base, and it rotted in place

Circular 174's swim vat was timber — 4×4 sills and posts, 2-inch plank walls — and period practice included **hybrid builds**: a concrete lower section for water-tightness with timber framing above grade. On this reading the concrete never extended beyond the base; the timber upper walls, cover leaves and splash boards simply **decayed where they stood** over a century. Nothing had to be hauled away. The **weathered timber post with wire standing beside the structure**, and the pipe rails, are exactly the surviving traces this reading predicts.

Which reading is true is decidable in one site visit: Reading A leaves buried concrete beyond the visible ends (probe finds it); Reading B leaves none (probe finds undisturbed soil). Either way the sediment inside the visible rectangle is the sampling target, and either way it must not be disturbed except under consent with an accredited laboratory.



Longitudinal section, to scale. Solid = observed; ghosted = interpreted from USDA Circular 183. New finds (timber post with wire, pipe rails) listed on the figure.

What would resolve it — in priority order

#	Measurement	Why it decides	Vat if…
1	Depth to hard concrete floor, probed through sediment at several points	The single most discriminating number. Vats are deep because cattle swim.	5–6.5 ft
2	Width at rim and again as low as reachable	A vat tapers; a ditch has parallel walls. Requires no digging.	narrows ~3 ft → ~1.5 ft
3	Photograph both ends separately	A vat is asymmetric: steep slide in, cleated ramp out. A ditch is uniform.	slide one end, ramp the other
4	Look for an adjacent flat concrete apron	The drip pen slopes back toward the vat and is unmistakable once seen.	apron 36 in × 20–40 ft
5	Total length rim to rim, plus GPS coordinates and a scale object in frame	Makes the record usable by anyone else and locatable again.	12–26 ft

Capacity check, once depth is known

Using the reported 15–20 ft × 3–4 ft footprint and the prismatoid volume the circulars themselves specify: a depth of 2 ft yields about 916 gal; 3 ft about 1,375 gal; 4 ft about 1,833 gal; 5.25 ft about 2,405 gal; 6.5 ft about 2,978 gal. The documented capacities are **1,470 gal** (Circular 183/207 concrete vat) and **2,088 gal** (Circular 174 swim vat). Depths at or above roughly 4 ft put the structure in the documented range; depths under 3 ft place it outside.

If it holds up

A suspected historic arsenical site is a regulatory matter with an established pathway. It should not be handled privately or by a contractor.

- **California DTSC** — the agency for suspected historic contamination. EnviroStor is their public system and shows no record for this area, which is itself a documented gap.
- **OC Public Works / Flood Control** — if the structure sits in or beside a county drainage easement, they hold the as-builts and the maintenance history.
- **Property owner consent** — LARMAC, the County, or Rancho Mission Viejo depending on parcel. This project's gate G10 already anticipates exactly this and requires consent plus an accredited laboratory.
- **Analyte suite:** arsenic *and lead*, plus DDT/DDE, toxaphene and PAHs. The lead ratio is what separates cattle-dip ground from orchard ground — dip ground carries arsenic without lead.

Statement class: field observation, unverified. **Provenance:** observer photograph and verbal dimensions, uncorroborated. Specifications quoted are A1 — USDA Bureau of Animal Industry Circulars 174 (1911), 183 (1911) and 207 (1912), held locally with full text and SHA-256 checksums.
This document does not assert that a dipping vat has been found, that arsenic is present at this or any location, that anyone has been exposed, or that any illness has any environmental cause. It records an observation and the measurements that would settle what it is. A single accredited laboratory result would outweigh everything in it.